

## PROJECT PROPOSAL

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# Campus Lost and Found

AI-Assisted Web Platform for Campus Item Recovery

### Software Engineering Project Proposal

Project Duration: 4 Months

Proposed Platform: Web Application

Primary Users: Students, Staff, Faculty, Admin

## Project Title

- Campus Lost and Found: AI-Assisted Web Platform for Campus Item Recovery

## Project Summary

- The proposed project is a secure web-based platform designed to help students, faculty, and staff report, track, match, and recover lost belongings inside a campus environment.
- The platform replaces informal recovery methods such as WhatsApp groups, notice boards, verbal communication, and manual registers with a centralized digital system.
- The system uses verified login, structured item reporting, AI-assisted matching, admin review, real-time notifications, and chat-based coordination.

### Existing Problems

- Students and staff often lose important items such as ID cards, phones, books, laptops, wallets, keys, and accessories.
- There is no single reliable platform where all lost and found cases are recorded and tracked.
- Current methods are scattered across WhatsApp groups, notice boards, security offices, and informal communication.
- Manual searching is time-consuming and depends on whether the right person sees the message at the right time.
- Found item details may be misused if they are made public, leading to false claims.
- Admins and security staff do not get a structured record of all active, matched, rejected, and resolved cases.
- There is no automatic system to compare lost reports with found reports using item details and uploaded images.

### Need for the Project

- A campus requires a secure, transparent, and organized lost-and-found platform that improves recovery speed and reduces misuse.

### Solution Description

- The proposed solution is a full-stack web application where only verified Thapar users can access the system.
- Users can submit lost item reports that are visible to campus users and found item reports that remain private until admin verification.
- The AI matching engine compares lost and found reports using text similarity, category, color, location, and uploaded image details.
- Suggested matches are sent to the admin dashboard, where the admin approves or rejects the match.
- After approval, the system notifies the owner and finder and enables real-time chat so they can coordinate the item return safely.

### Professional Value

- The solution improves trust because the final match is not decided only by the system; it is verified by an authorized admin.
- It improves efficiency because users do not have to manually search through many informal messages.
- It improves accountability because every report is linked to a verified institutional email ID.

### Primary Objectives

- Develop a centralized platform for reporting, tracking, matching, and resolving campus lost-and-found cases.
- Allow only verified Thapar users to register and access the platform.
- Provide separate user and admin roles with proper access permissions.
- Enable users to report lost items with details such as item name, category, description, color, location, date, and image.
- Enable users to report found items while keeping sensitive found item details private from the public.
- Use AI/NLP methods to automatically suggest possible matches between lost and found reports.
- Use admin approval to verify ownership and reduce false claims.
- Provide real-time notifications for match updates, admin decisions, and chat messages.
- Provide real-time chat only after a match has been approved by the admin.
- Maintain digital records for active, matched, rejected, and resolved cases.

### Frontend

- **React.js:** Used to build a responsive and interactive user interface.
- **Vite:** Used as a fast development and build tool for the React application.
- **HTML, CSS, JavaScript:** Used for structure, styling, and client-side interaction.
- **Responsive Design:** Ensures that the platform works properly on desktop and mobile browsers.

### Backend

- **Node.js:** Used for server-side execution and backend logic.
- **Express.js:** Used to build REST APIs for login, item reports, search, matching, admin approval, notifications, and chat.
- **JWT:** Used for secure user sessions and protected routes.
- **Socket.IO:** Used for real-time chat and instant notification delivery.

### Database and Cloud Services

- **Firebase Authentication:** Used to manage user login and identity verification.
- **Firestore:** Used as a NoSQL database for users, items, matches, messages, notifications, and status records.
- **Firebase Storage or Cloud Image Storage:** Used to store uploaded item images securely.

### AI Matching Approach

- **NLP:** Used to process item descriptions entered by users.
- **TF-IDF Vectorization:** Converts item descriptions into numerical form so that they can be compared.
- **Cosine Similarity:** Calculates how similar two item descriptions are.
- **Keyword Matching:** Checks important words such as brand name, item type, color, model, and unique marks.
- **Category Matching:** Compares item categories such as ID card, electronics, books, wallet, bottle, keys, and accessories.
- **Location Matching:** Gives a higher match score when lost and found locations are the same or nearby.
- **Color Matching:** Compares user-entered color details or color information extracted from images.
- **OCR:** Can be used to read visible text from uploaded images, such as ID card text, book titles, labels, or names.
- **Image Classification:** Can identify the object type from the uploaded photo.
- **CNN or CLIP-style Models:** Can compare image features with item descriptions to improve matching accuracy.

### Final Match Score

- The final score can be calculated by combining description similarity, category match, location match, color match, and image similarity.

## 6. AI, NLP, and Image Matching

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- If the score is above the selected threshold, the system sends the match suggestion to the admin for verification.

### Thapar Email Verification

- The system will allow registration and login only through official Thapar email IDs.
- The email domain will be checked during signup/login. Example accepted format: username@thapar.edu.
- If the email does not belong to the accepted Thapar domain, the system will deny access.
- Firebase Authentication will verify the user credentials.
- After successful login, JWT will be generated to maintain a secure session.
- Role-based access will separate normal users from admin users.
- Students and staff can report and search items, while admins can verify matches and manage records.

### Main Modules

- **User Authentication Module:** Handles signup, login, logout, Thapar email verification, JWT sessions, and role-based access.
- **Lost Item Reporting Module:** Allows users to create public lost item reports with description, category, color, location, date, and image.
- **Found Item Reporting Module:** Allows users to submit found item reports that remain private until admin verification.
- **Search and Filter Module:** Allows users to search lost items using category, location, date, and keywords.
- **AI Matching Module:** Compares lost and found reports using NLP, similarity scoring, location, color, category, and image details.
- **Admin Approval Module:** Allows admins to review AI-suggested matches and approve or reject them.
- **Notification Module:** Sends alerts for match suggestions, admin decisions, chat messages, and item status updates.
- **Chat Module:** Allows real-time communication between the owner and finder after admin approval.
- **User Dashboard Module:** Shows user posts, current status, notifications, and matched results.
- **Admin Dashboard Module:** Allows admins to manage reports, matches, users, and resolved cases.

Development Timeline

|         |                         |                                                                                                                                                                                                          |
|---------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Month 1 | Planning and Foundation | Finalize scope, prepare SRS, define users, prepare database design, create GitHub repository, set up React frontend, Node/Express backend, and Firebase project.                                         |
| Month 2 | Core User Features      | Implement Thapar email login, role-based access, lost item reporting, found item reporting, image upload, user dashboard, search, filters, and basic item status tracking.                               |
| Month 3 | AI and Admin Features   | Implement NLP matching using TF-IDF/cosine similarity, add category/color/location scoring, add OCR or image-based support, build admin dashboard, approval workflow, notifications, and real-time chat. |
| Month 4 | Testing and Submission  | Perform functional testing, security testing, role testing, responsive UI testing, bug fixing, performance improvements, final documentation, screenshots, README, presentation, and demo preparation.   |

## 10. Expected Outcomes and Conclusion

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### Expected Outcomes

- A complete working web application for campus lost and found management.
- Verified access using official Thapar email IDs.
- Secure reporting of lost and found items.
- Private handling of found item details to reduce fake claims.
- AI/NLP-based matching between lost and found reports.
- Support for image-based item identification and OCR-based text detection.
- Admin approval workflow for match verification.
- Real-time notifications and chat functionality.
- User and admin dashboards for proper tracking and management.
- Improved recovery speed, transparency, and accountability.

### Conclusion

- Campus Lost and Found is a practical software engineering project that addresses a real campus problem using modern web technologies and AI-assisted matching.
- The project is feasible within a four-month academic timeline and can later be extended with a mobile app, QR-based item tagging, advanced analytics, and integration with campus security systems.